

BASE ORCADAS, Antarctica

WMO: 889680

Lat: 60.7378S

Lon: 44.7383W

Elev: 26

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-15.6	-11.3	-23.1	1.5	-15.1	-17.4	2.1	-11.0	48.1	23.1	46.4	22.8	12.1	140	1.077

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.1	42.3	38.8	40.5	37.5	39.1	36.5	39.4	41.5	38.1	39.8	37.0	38.4	16.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
37.6	33.1	39.5	36.5	31.8	38.3	35.7	30.7	37.3	15.1	41.7	14.5	40.1	14.0	38.7	49.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
42.7	33.7	29.5	-20.5	47.3	3.8	2.7	-23.2	49.2	-25.4	50.8	-27.5	52.3	-30.3	54.3		
			-20.6	43.2	3.5	2.0	-23.1	44.7	-25.1	45.8	-27.1	46.9	-29.6	48.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	34.4	34.3	32.9	28.8	23.5	19.3	14.7	16.5	21.2	27.7	31.2	33.2
	DBStd	10.68	2.45	2.68	3.77	5.79	9.37	10.97	13.46	12.57	10.77	7.01	4.18	2.57
	HDD50	8605	484	439	529	637	822	921	1094	1037	865	692	565	520
	HDD65	14080	949	859	994	1087	1287	1371	1559	1502	1315	1157	1015	985
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.5	13.0	14.3	15.3	15.0	14.8	14.1	14.6	15.0	14.9	15.2	14.2	13.4
Precipitation	PrecAvg	31.6	2.8	3.0	4.1	3.3	2.4	1.9	2.0	2.2	2.3	2.9	2.6	2.2
	PrecMax	39.3	4.6	4.8	7.0	4.9	5.1	3.1	3.8	4.1	3.9	5.7	4.6	3.8
	PrecMin	23.0	1.8	1.9	2.5	1.8	1.0	0.9	0.6	0.4	0.8	0.8	1.0	0.6
	PrecStd	4.3	0.7	0.6	1.0	0.9	0.9	0.7	0.9	1.1	0.9	1.0	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	44.1	44.5	41.4	39.8	37.1	36.0	37.0	40.4	40.5	42.5	43.2
		MCWB	40.2	40.5	40.8	38.1	36.6	34.7	33.3	34.3	37.1	37.0	38.5	38.9
	2%	DB	40.9	41.1	41.1	37.8	36.5	34.1	33.5	34.2	36.6	38.0	39.7	39.8
		MCWB	37.7	38.4	38.3	35.5	34.5	32.7	32.3	32.8	34.3	35.2	36.4	36.6
	5%	DB	39.1	39.2	39.1	36.0	34.8	32.7	32.2	32.7	34.5	36.1	37.6	37.8
		MCWB	36.7	36.9	37.1	34.5	33.5	31.6	31.4	31.7	32.9	33.9	35.0	35.2
	10%	DB	37.7	37.8	37.5	35.0	33.5	31.5	31.2	31.5	33.0	34.6	36.0	36.6
		MCWB	35.8	36.0	35.9	33.7	32.6	30.6	30.5	30.7	31.9	33.0	34.0	34.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.1	41.3	41.0	38.6	37.0	34.9	34.1	34.7	37.6	37.6	39.2	39.4
		MCDB	43.4	43.4	44.1	40.4	38.8	36.4	34.9	35.9	39.5	39.6	41.6	42.3
	2%	WB	38.4	38.7	38.9	36.0	34.9	33.1	32.4	33.1	34.7	35.6	36.8	37.1
		MCDB	40.1	40.4	40.4	37.1	35.9	33.8	33.0	34.0	36.1	37.5	39.1	39.2
	5%	WB	37.1	37.5	37.5	35.0	33.8	32.0	31.6	31.9	33.1	34.2	35.4	35.9
		MCDB	38.4	38.7	38.7	35.8	34.6	32.6	32.1	32.6	34.0	35.5	37.0	37.3
	10%	WB	36.2	36.5	36.2	34.0	32.7	30.8	30.6	30.9	32.0	33.2	34.3	34.9
		MCDB	37.4	37.6	37.2	34.8	33.4	31.4	31.3	31.5	32.8	34.2	35.6	36.2
Mean Daily Temperature Range	5% DB	MDBR	5.1	4.7	5.4	6.1	9.0	10.9	13.2	12.7	11.8	8.4	7.0	5.5
		MCDBR	7.4	7.1	7.5	6.7	7.7	9.1	10.9	10.5	9.8	8.5	8.8	7.7
		MCWBR	5.4	5.6	5.9	5.7	6.7	8.7	10.5	10.0	8.5	6.9	6.8	5.5
	5% WB	MCDBR	6.7	6.6	7.2	6.4	7.1	8.8	10.7	10.1	9.4	7.8	8.2	7.1
		MCWBR	5.3	5.6	6.0	5.8	6.4	8.7	10.6	10.0	8.5	6.7	6.7	5.5
Clear-Sky Solar Irradiance	taub	0.316	0.314	0.302	0.284	0.229	0.179	0.209	0.251	0.277	0.302	0.295	0.305	
	taud	2.497	2.524	2.527	2.437	2.193	1.991	2.118	2.270	2.326	2.349	2.429	2.465	
	Ebn at Noon	306	292	270	224	165	129	167	229	271	291	311	313	
	Edh at Noon	31	28	23	17	11	8	11	19	27	32	33	33	
All-Sky Solar Radiation	RadAvg	1428	1009	579	290	109	51	84	288	691	1186	1652	1722	
	RadStd	104	88	29	30	14	4	5	27	63	141	206	175	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page